

# Operator Manual – "AutoGelato" Production System

## 1. Pre-Shift Setup:

- Ensure that initial inventory levels are loaded into the designated registers: Raspberry (**D1200**), Lemon (**D1202**), Stabilizer (**D1204**), and Gummy Bears (**D1206**).
- Verify that the Emergency Stop button (**X5**) is released (pulled out).

## 2. Starting the Workday:

- To initialize the system, press the X7 switch.
- Indicator: The Main Operation Light (Y7) will turn on and remain illuminated throughout the day.

## 3. Phase A: Ice Cream Preparation (Switch X1):

- Set the desired package type using the Analog Input 1 potentiometer:
  - Standard Package: Value between 0 and 2000.
  - Premium ("Pinuk") Package: Value above 2400.
- Automatic Check: The system will verify raw material availability. If materials are insufficient, the system will shut down (see "Edge Cases").
- Indicator: The Preparation Light (Y0) will illuminate for 5 seconds.

## 4. Phase B: Cooling Process (Switch X2):

- After the Preparation Light turns off, flip the X2 switch.
- The ice cream mixture is transferred to the cooling tanks.
- Indicator: The Cooling Light (Y2) will illuminate. The phase ends automatically upon reaching the required temperature or time limit.

## 5. Phase C: Automatic Mixing & Packaging (System Feedback):

- This phase is fully automated. The operator does not need to toggle a switch, however, you must monitor the following indicators to confirm the process:
  - Mixing Indicator: The Mixing Light (**Y2**) will turn on during the process.

- Package Type Feedback:
  - **Standard Package:** Lights **Y5** and **Y6** will flash alternately for 3 seconds.
  - **Premium Package:** Lights **Y5** and **Y6** will stay on steadily for 3 seconds.
- The system will automatically update the gummy bear inventory and the total unit counter once the cycle finishes.

## **6. End of Day / Automatic Shutdown:**

- The system will automatically shut down if a shortage of raw materials is detected.
- At the end of the day, the total count of produced units will be displayed via the Analog Output (D8282).

## **7. Emergency Procedures:**

- In case of any malfunction or safety hazard, press the Emergency Stop (X5).
- This will immediately reset all process variables and turn off the Main Operation Light.

## **Operator Notes:**

- Sequence Control: Do not activate a switch for the next phase until the current indicator light has turned off.
- Invalid Input: If the "Invalid Input" flag (M14) is triggered, check the Analog Input 1 calibration to ensure it falls within the permitted ranges.